Conduction, convection, radiation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify mechanisms of heat transfer and support each classification.

SUCCESS CRITERIA - I can...

1. Describe contact transfer, bulk fluid motion and electromagnetic transfer.
2. Use evidence from the situation.
3. Recognize that more than one mechanism can act.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “conduction” and “convection” to explain your model or result. Sentence starters are optional.

conduction: energy transfer through microscopic interactions in matter

convection: energy transfer through bulk motion of a fluid

radiation: energy transfer by electromagnetic waves

01 Warm-up: bring back an earlier idea

A device transfers 15 J in 3 s, then performs the same transfer in 6 s.
Find both average powers and explain the difference.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $E_i + E_{in} = E_f + E_{out}$.
Account for energy across the system boundary. Conduction: matter interactions; convection: fluid motion; radiation: electromagnetic waves.

Worked reasoning: A metal rod transfers energy from its hot end toward its cool end by conduction. A circulating fluid can carry energy by convection; radiation can cross a vacuum.

Your turn: Draw one distinct, labeled model for each mechanism.
Show the direction of energy transfer.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Conduction	Convection	Radiation
Sketch:	Sketch:	Sketch:
_____	_____	_____

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Classify: a metal spoon warming in soup; circulating water in a pot; sunlight warming your hand.
Give a reason for each.

Sentence starter: I notice … . This supports … because … .

Situation	Mechanism	Evidence
Spoon in soup	____	____
Circulating water	____	____
Sunlight on skin	____	____

Marzano check - circle: 1 2 3

04 Apply the physics

A pot of water is heated from below. On the model, draw a circulation loop: warmer, less-dense water rising and cooler water sinking.
Explain how this bulk motion differs from energy transfer along a solid metal spoon.

Sentence starter: Both … . They differ in … because … .



Heater below pot

Marzano check - circle: 1 2 3

05 Explain your reasoning

An asteroid travels through nearly empty space. Which of these mechanisms can transfer thermal energy to it from sunlight? Why not ordinary air convection?

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A rolling object loses 24 J gravitational energy but gains only 16 J translational kinetic energy.
Name two possible destinations of the difference.
Show the energy changes and transfers; label units.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Explain how energy can travel from the Sun to Earth through space.

Sentence starter: My claim is … . The diagram/data show …, so … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Design insulation for a hot cup using one change for each transfer mechanism.
Explain each change.

Sentence starter: My design shows … . It meets the requirement by … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3